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IMMOBILIZATION OF JACK BEAN UREASE ON HYDROXYAPATITE: UREASE IMMOBILIZATION IN ALKALINE SOILS

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Summary—Jack bean urease was adsorbed and immobilized on hydroxyapatite, a stable calcium phosphate mineral commonly found in alkaline soils. The stability and kinetic properties of the resulting complex were investigated to derive a model that explains the stabilisation of urease in alkaline soil. The adsorption isotherms revealed: (1) an increase of the affinity of urease for hydroxyapatite with increasing ionic strength at constant pH and (2) a decrease of affinity with increasing pH at constant salt concentration. The optimum pH for enzymatic activity increased from 7 to 8 upon adsorption. The Michaelis–Menten parameters for free urease ($V_{max} = 230.7$ U mg^{-1}; $K_m = 7.45$ mM) and for immobilized urease ($V_{max} = 152.9$ U mg^{-1}; $K_m = 6.89$ mM) showed a moderate decrease of enzyme specific activity, but little change of substrate affinity. Urease adsorption resulted in a 2-fold increase in enzyme stability with time and a 7-fold increase in resistance to proteolytic hydrolysis. The results presented indicate that hydroxyapatite could have a significant role in the stabilisation of extracellular urease in alkaline soils. © 1998 Elsevier Science Ltd. All rights reserved

INTRODUCTION

Urease (urea amydohydrolase, E.C.3.5.1.5.) hydrolyses urea, yielding ammonia and carbonate (Andrews *et al.*, 1989). The presence of large amounts of urease in soils allows the worldwide use of urea as a nitrogen fertilizer. Soil urease originates mainly from plants (Polacco, 1977) and microorganisms (Mobley and Hausinger, 1989) and is found both as intra- and extra-cellular enzyme (Burns, 1986). It has been reported that extracellular urease, associated with soil organo–mineral complexes, is more stable than urease in the soil solution (Burns, 1986) and that humus–urease complexes extracted from soil are highly resistant to denaturing agents such as extreme temperatures and proteolytic attack (Nannipieri *et al.*, 1978). On the other hand, urease extracted from plants or microorganisms is rapidly degraded in soil by proteolytic enzymes (Burns *et al.*, 1972a; Pettit *et al.*, 1976; Zantua and Bremner, 1977). These experimental results appear to be consistent with the hypothesis that a significant fraction of ureolytic activity in soil is carried out by extracellular urease, which is stabilized by immobilization on organic and mineral soil colloids. Complexes of urease with a wide range of soil components have been prepared to investigate adsorption, kinetic and stability properties of the immobilized urease in soil. Most of these were prepared using urease from jack bean (*Canavalia ensiformis*), a plant belonging to the family of Leguminosae. Jack bean urease is the most studied and the best known among the ureases (Bremner and Mulvaney, 1978). Consequently, it is the most widely used urease for immobilization on soil components: it has been immobilized on clay (Gianfreda *et al.*, 1992; Lai and Tabatabai, 1992), humic or humic-like material (Nannipieri *et al.*, 1978; Burns, 1986; Vaughan and Ord, 1991), aluminium hydroxide (Gianfreda *et al.*, 1992) and organo–mineral complexes (Boyd and Mortland, 1985; Lai and Tabatabai, 1992; Gianfreda *et al.*, 1995). All of these studies have shown that the carrier used in the immobilization procedure can differently affect the urease kinetic and stability properties.

Despite the number of soil constituents investigated as urease carriers, no attention has been given so far to calcium phosphates, which represent an important component of alkaline and calcareous soils. Fertilizer phosphorus is precipitated as calcium phosphate in the presence of Ca^{2+} ions and progressively transformed from metastable, amorphous forms to stable, insoluble phases (Matar *et al.*, 1992). Among these, hydroxyapatite (APA), $Ca_{10}(PO_4)_6OH_2$, is reported to be an important constituent of sediments and soils and also to have a great affinity for organic molecules such as amino

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acids (Aoba and Moreno, 1985) and humic, fulvic and tannic acids (Inskeep and Silvertooth, 1988). These properties suggest that APA could play an important role in enzyme adsorption and immobilization of soil urease. We report the results of a study of urease adsorption on APA and propose a possible model for accumulation of urease in alkaline soils.

MATERIALS AND METHODS

Chemicals

Highly purified urease from jack bean (*Canavalia ensiformis*) and hydroxyapatite were purchased from Sigma Chemical Co., St. Louis, MO. Pronase was purchased from Boehringer Mannheim Italia SpA.

Formation of the APA–urease system

APA (10 mg) was equilibrated for 10 min with 10 ml of buffer (2 mM Tris·HCl, 2 mM Na_2SO_3, 10 mM NaCl) using a oscillating shaker (50 cycles · min^{-1}) at 25°C. The suspension obtained was centrifuged for 15 min at $27{,}000 \times g$ at 25°C and the supernatant was discarded. The APA pellet was resuspended in the buffer containing 250 μg of urease and the resulting suspension was shaken (50 cycles min^{-1}) for 30 min at 25°C. The pellet obtained after centrifugation (15 min $27{,}000 \times g$, at 25°C) was washed thoroughly with buffer, using two cycles of resuspension–centrifugation. After this treatment, no further urease activity was detected in the supernatant. The pellet (APA–urease) obtained after the last centrifugation step was stored at 4°C overnight and utilized for subsequent experiments. The storage did not determine any variation in APA–urease pellet activity.

The addition of 2 mM sodium sulphite and 10 mM NaCl to the Tris·HCl buffer improves the stability of urease (Blakeley and Zerner, 1984) and the accuracy of urease activity assays. The urease activity in this buffer showed no variation over a period of 16 h at 25°C. The pH of buffer was varied using 0.5 M HCl or 0.1 M NaOH.

Assay of urease activity

An amount of urease equal to 250 μg was added to all samples assayed. Triplicate measurements of urease activity were performed, on both free and adsorbed enzyme, with a pH-stat method using a Crison Compact Titrator 2000 pH-meter and a Crison 52-20 electrode. The activity was measured by recording the volume of a 10 mM HCl solution necessary to maintain pH = 8.0 of a solution of free enzyme, or a suspension of APA–urease, in the unit time. The activity measurements were carried out after linearity of the addition curve was reached, i. e. after 1 min, in order to allow uniform urea concentration in the sample. Measurements were performed in 10 ml of the buffer containing 400 mM urea.

The pH activity profiles for free and adsorbed urease were obtained by changing the pH of the buffer in the range 6.5–9.0 and carrying out the pH-stat method at the corresponding value of pH. One unit of enzyme activity is defined as the amount of enzyme needed to hydrolyze 1 μmol urea min^{-1} at 30°C. The calculation of urease activity units was carried out taking into consideration the pH of samples, the volumes of HCl added and the acidity constants of the products of urea hydrolysis, using the procedure of Blakeley *et al.* (1969).

Adsorption isotherms

Buffer (2.0 ml) containing urease amounts ranging from 0.25 to 1 mg, were added to 20 mg APA, previously equilibrated with the same buffer and the adsorption procedure was performed as described above. The protein content in the equilibrium solutions was determined using the BioRad Protein Assay kit. The amount of urease adsorbed on APA was calculated by difference. Protein concentrations in the equilibrium solution were used to plot equilibrium adsorption isotherms. Adsorption isotherms were measured using the buffer at pH 7, 8 and 9 and in buffer at pH 8.0 with 0–50 mM NaCl.

The experimental adsorption data were interpreted using the simplest adsorption model, assuming (i) that the adsorbed molecules form a single layer on the solid surface and (ii) that protein–protein interactions are absent. The model can be analyzed using the Langmuir equation:

$$x/m = KCb/(1 + KC)$$

where x is the amount of urease adsorbed (μg), m is the amount of APA utilized (mg), K is a constant related to the binding energy, b is the maximum adsorption of urease (μg urease mg^{-1} APA) and C is the equilibrium concentration of the enzyme (μg ml^{-1}). The experimental data were fitted to the Langmuir equation using a non-linear steepest descent curve fitting algorithm, implemented in the Mac Curve Fit 1.2 program.

Measurement of K_m and V_{max} for free and adsorbed urease

Kinetic experiments were performed, for both free and adsorbed urease, at pH 8. The activity was measured as a function of substrate concentration and K_m and V_{max} were determined by fitting the experimental data to the Michaelis–Menten equation using a non-linear steepest descent curve fitting algorithm, implemented in the Mac Curve Fit 1.2 program.

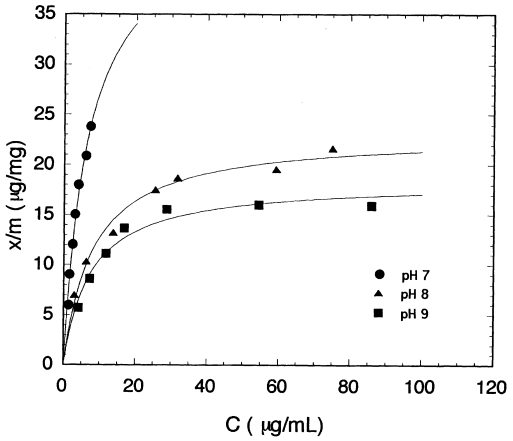

Fig. 1. Isotherms of adsorption of jack bean urease on hydroxyapatite. Effect of pH (μg urease ml^{-1} in equilibrium solution vs μg urease mg^{-1} APA).

Stability of free and adsorbed urease with time and in the presence of pronase

The stability of free and adsorbed urease with time was monitored at 25°C and pH 8 by assaying the activity after 1, 2, 3 and 4 d. The stability of free and adsorbed forms of urease was also tested in the presence of pronase, using buffer solutions containing 0.013 pronase units ml^{-1}. One unit of pronase activity is defined as the amount of enzyme capable of liberating, in 1 min, Folin-positive amino acids and peptides corresponding to 1 mmol tyrosine.

RESULTS AND DISCUSSION

Adsorption of urease on hydroxyapatite: adsorption isotherms

Figure 1 shows isotherms of adsorption of urease on APA, at varying pH. It is evident that the *b* value in the Langmuir isotherm is considerably reduced as pH increases from 7 to 9 (Table 1). This behavior suggests that adsorption is at least partially due to interactions between pH-dependent opposite charges of both urease and APA surfaces. Urease has an isoelectric point of 4.9 (Reithel, 1971) and a pH-dependent negative net charge in the pH range considered. On the other hand, the interruption of the crystal structure of APA provides both positive (Ca^{2+}) and negative (oxide, hydroxide and phosphates) pH-dependent charges at its surface. The reduction of enzyme adsorption with increasing pH can then be explained by an increase of the number of negative charged sites on both surfaces, increasing repulsive and decreasing attractive interactions. It is also possible that hydroxide ions in solution compete with urease *R*–C(O)O$^-$ groups for Ca^{2+} ions on the APA surface, consequently reducing urease adsorption. The COO^-/Ca^{2+} interactions have been considered responsible for the adsorption of amino acids (Kresak *et al.*, 1977; Aoba and Moreno, 1985) and fulvic and humic acids (Inskeep and Silvertooth, 1988) on APA. The role of Ca^{2+} ions was also proposed to be important in a study of the immobilization of urease on Ca–polygalacturonate (Ciurli *et al.*, 1996). The Langmuir parameter *K* is not affected by pH (Table 1). This can be explained by assuming that no variation occurs in the chemical nature of the adsorption sites, which are simply reduced in number as pH increases.

Table 1. Dependance of Langmuir parameters on pH

pH (S.D.)	*K* (ml μg^{-1})	*b* (μg mg^{-1})
7	6.65 (1.37)	45.36 (5.55)
8	8.16 (1.15)	23.04 (0.85)
9	7.51 (1.21)	18.33 (0.81)

Table 2. Dependance of Langmuir parameters on ionic strength

μ (S.D.) (mM)	*K* (ml μg^{-1})	*b* (μg mg^{-1})
0	5.41 (0.50)	19.92 (0.41)
10	8.34 (1.52)	22.84 (1.09)
50	36.49 (4.55)	55.73 (4.31)

Both Langmuir parameters, *b* and *K*, increase with NaCl concentration (Table 2). This behavior can be explained by the higher ionic strength reducing both the attractive and the repulsive (coulombic) interactions between surface charges by increasing the dielectric constant of the solution, thus revealing the existence of strong hydrophobic attractive interactions (Fig. 2).

pH-dependence

Each enzyme is characterized by an optimum pH value, at which maximum activity is measured. The pH–activity relationships of both free and adsorbed urease has been studied at pH 6.5 to 9.0. Figure 3 shows that free urease activity has a maximum at pH 7.0, while adsorbed urease is most active at pH 8.0. Similar results were obtained in other studies (McLaren and Parker, 1970; Katchalski *et al.*, 1971)

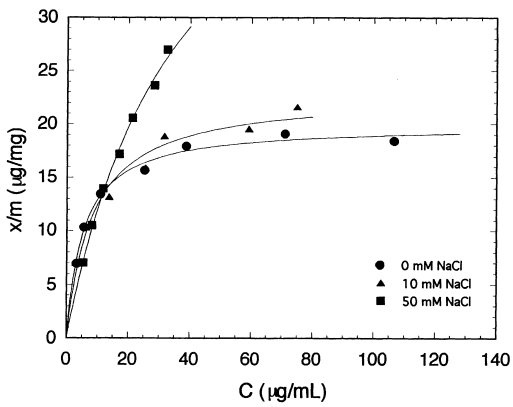

Fig. 2. Isotherms of adsorption of jack bean urease on hydroxyapatite. Effect of ionic strength (μg urease ml^{-1} in equilibrium solution vs μg urease mg^{-1} APA).

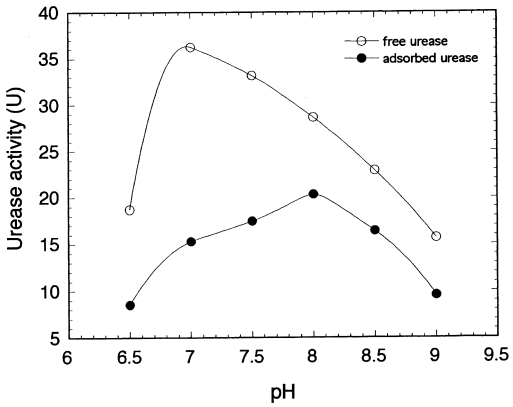

Fig. 3. Effect of pH on the activity of free and adsorbed jack bean urease.

in which clay was used as a carrier for different enzymes: the pH optimum of an adsorbed enzyme was 1 or 2 units higher than that of the free enzyme, a common character for all immobilized enzymes on anionic carriers. The size of this shift depended on the type of clay used, its cation exchange capacity and hydration level and could be explained by considering that the clay surface has a Brönsted acidity considerably higher than the bulk solution (Boyd and Mortland, 1985). In contrast, other studies have pointed out no significant variations in the pH optimum of urease immobilized on different soil components (Boyd and Mortland, 1985; Gianfreda *et al.*, 1992, 1995; Ciurli *et al.*, 1996).

Kinetics

The Michaelis–Menten parameters for free urease, $V_{max} = 230.7$ U mg^{-1} ($\pm$6.2) and $K_m = 7.4$ mM ($\pm$0.7) and for immobilized urease, $V_{max} = 152.9$ U mg^{-1} ($\pm$5.5) and $K_m = 6.9$ mM ($\pm$0.88), show a 34% reduction of V_{max} compared to the free urease, while no variation in the K_m is observed (Fig. 4). These K_m values are comparable to values reported by other authors (Bremner and Mulvaney, 1978; Gianfreda *et al.*, 1992). The decrease of V_{max} can be explained by assuming either that the adsorption on APA affects the tertiary structure of the enzyme thus reducing its catalytic activity, or that a fraction of enzyme molecules is inactivated by the adsorption process. On the other hand, the invariance of K_m suggests that no interaction occurs between urea and APA, so that the concentration of substrate necessary for the maximum reaction rate to occur is the same for free and adsorbed urease. To ascertain this, an adsorption procedure of a 400 mM urea solution (corresponding to the substrate concentration used in the urease assay) on APA was carried out, in the same conditions adopted for urease immobilization. The amount of urea in the resulting equilibrium solution (393 ($\pm$10) mM), assayed using a diacetyl monoxime method (Mulvaney and Bremner, 1979) confirmed that urea is not adsorbed on APA.

Several studies on enzyme immobilization have reported that both specific activity and substrate affinity are considerably reduced after adsorption of enzymes on a carrier (Burns, 1986), in contrast with our results. However, jack bean urease was shown not to strictly follow this general behavior. No variation was actually found in the kinetic parameters of a jack bean urease adsorbed on a clay–organic complex (Boyd and Mortland, 1985). A decrease of

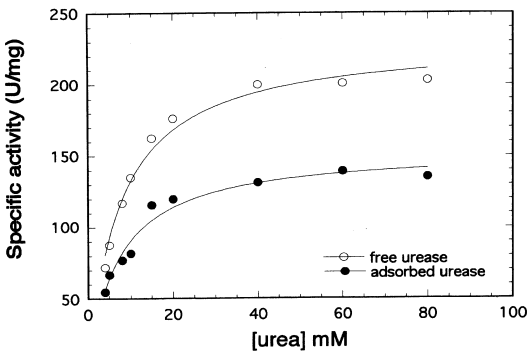

Fig. 4. Michaelis–Menten plots for free and adsorbed jack bean urease. (Concentration of urea vs units of urease activity mg^{-1} protein).

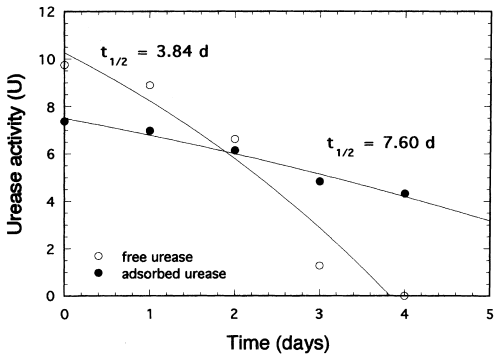

Fig. 5. Stability with time of free and adsorbed jack bean urease.

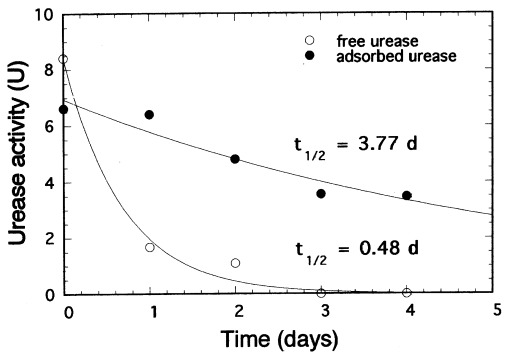

Fig. 6. Stability with time of free and adsorbed jack bean urease in the presence of pronase.

K_m and a reduction of the specific activity of a jack bean urease after immobilization on montmorillonite, aluminium hydroxide and a montmorillonite–aluminium hydroxide complex have been reported (Gianfreda *et al.*, 1992).

Stability properties

Figure 5 and 6 illustrate the stability of free and adsorbed urease with time and in the presence of pronase, respectively. In the first case, the enzyme activity after 4 d was 60% of the original value for the adsorbed urease, whereas free urease was completely inactive. After 4 d in the presence of pronase, adsorbed urease retained 50% of its original activity, while free urease showed an activity rapidly decreasing to zero after 3 d. Moreover, the stability of adsorbed urease in the presence of pronase was greater than that of free urease in the absence of pronase. The values of the calculated half-lives of free and adsorbed urease, in the absence and in presence of pronase respectively (Figs 5 and 6), indicate that immobilization causes a 2-fold stability increase with time and a 7-fold stability increase in the presence of pronase, confirming the hypothesis of an enzyme-protection mechanism by APA in soil. The half-lives of free and adsorbed urease were calculated by fitting the experimental data to an exponential function.

In general, enzymes accumulated in soil show a high stability to proteolysis. This property is thought to be one of the most important aspects of the protection mechanism provided by soil (Burns *et al.*, 1972b; Nannipieri *et al.*, 1974; Pettit *et al.*, 1976). The immobilization of jack bean urease on solid phases commonly found in soil under laboratory conditions has given results not always consistent with a stabilization effect upon adsorption, as we observed in this study. Investigations on complexes of jack bean urease and expandable and not-expandable clays (Burns *et al.*, 1972a; Gianfreda *et al.*, 1992, 1995), Al-hydroxide (Gianfreda *et al.*, 1992), tannic acid and hexadecyl-trimethylammonium (Boyd and Mortland, 1985) showed that these substances are not able to protect the enzyme. Proteolysis is partially decreased by adding lignin to a urease–bentonite complex (Burns *et al.*, 1972a).

Our results indicate that hydroxyapatite could represent an efficient carrier of extracellular urease in alkaline soils.

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